Cambridge (CIE) A Level Computer Science

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Algorithms

Contents

- Algorithm Basics
- Pseudocode
- Translation Skills
- Refinement & Logic

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Definition of algorithm

What is an algorithm?

- An algorithm is a **solution to a problem** expressed as a **sequence of defined steps**
- Methods of writing an algorithm before attempting to program a solution include:
  - **Structured English**
  - **Pseudocode**
  - **Flowcharts**

Structured English

- Structured English is a **human-readable method** for describing algorithms using a combination of **natural English language** and **programming logic**
  - It uses clear English phrases to describe each step
  - Logic structures like **IF...THEN**, **REPEAT**, and **WHILE** may appear, but without strict syntax rules
  - Often used in the early planning stages before converting to pseudocode

```
Ask the user to enter their age
If the age is 18 or over
→ Display a welcome message
Otherwise
→ Display an access denied message
```

Pseudocode

- Pseudocode is a **precise, structured, and language-independent way** of describing an algorithm that resembles a programming language
- It **follows specific exam-board-defined syntax**
  - It includes formal elements like IF, THEN, ELSE, WHILE, REPEAT, DECLARE, and ← for assignment
  - Students must follow **CIE's pseudocode format** in exams

```
INPUT Age
IF Age >= 18 THEN
   OUTPUT "Welcome to the site"
ELSE
   OUTPUT "Sorry, this site is for users 18 and over"
ENDIF
```

Flowcharts

- Flowcharts are a **visual tool** that uses **shapes to represent different functions** to **describe an algorithm**

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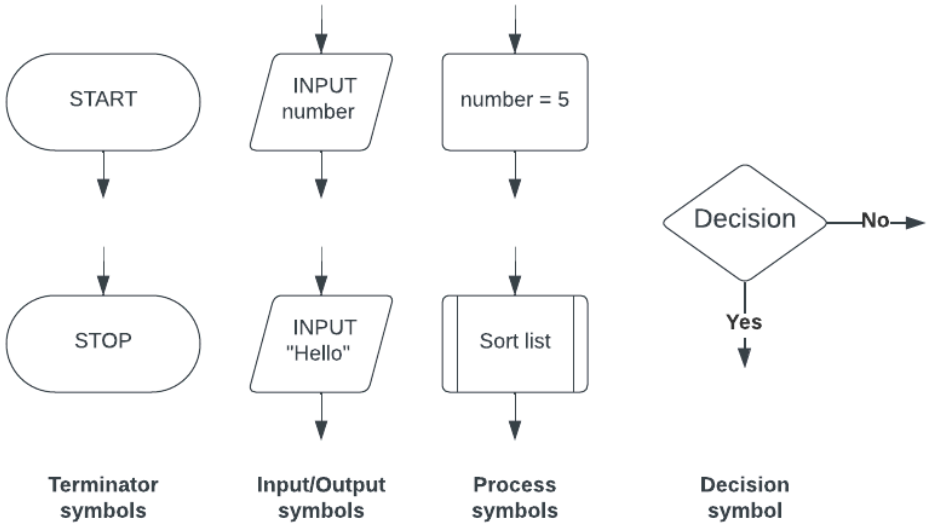
- Used to **visualise the flow of control** in a system
- Standard symbols include:
  - **Oval** for Start/End
  - **Parallelogram** for Input/Output
  - **Rectangle** for Processes
  - **Diamond** for Decisions
- Arrows show the **sequence of operations**



Identifier tables

What is an identifier table?

- An identifier table is used when writing **pseudocode** to keep track of all the **identifier names** used in an algorithm
- An **identifier** is the **name given to a variable, constant, array, procedure**, or any other named element in the pseudocode

Why use an identifier table?

- It helps you stay **organised** when designing an algorithm
- Ensures **consistent naming**
- Makes it easier to understand what each identifier stores or does
- Useful in **exam questions** when you are asked to declare variables clearly

Identifier naming rules

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- Must start with a **letter** (A–Z or a–z)
- Can include **letters**, **digits (0–9)**, and **underscores (_)**
- **Accented characters** and **symbols** are not allowed
- Identifiers are **case insensitive** (e.g. Total and total are treated the same)

Identifier	Description
StudentName	Stores a student's full name
TestScore	Holds a test score value
MAX_SCORE	Maximum score allowed
FormList	Stores names in a form group

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Input, process, output

What is an input?

- An input is **data or information being entered/taken into a program** before it is processed in the algorithm
- An input can come from a variety of sources, such as:
  - **User** - keyboard, mouse, controller, microphone
  - **Sensors** - temperature, pressure, movement
- Values are input using the INPUT command

```
INPUT <identifier>
```

What is a process?

- A process is **a doing action performed in the algorithm** that transforms inputs into the desired output. The central processing unit (**CPU**) executes the instructions that define the process
- An example would be:
  - **Comparing two numbers**
  - **Calculating an average**

What is an output?

- An output is **the result of the processing in an algorithm** and usually the way a user can see if an algorithm works as intended
- An output can take various forms, such as:
  - **Numbers** - result of calculations
  - **Text**
  - **Images**
  - **Actions** - triggering events
- Values are output using the OUTPUT command

```
OUTPUT <value(s)>
```

More than one value can be output in the same command when separated by commas

```
OUTPUT "First name: ", Fname, "Surname: ", Sname
```

Example 1 - Area of a shape

- A user wants to write a program to calculate the area of a shape

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Input	Process	Output
▪ Length ▪ Width	▪ Length X width	▪ Area

```
// Calculate the area of a rectangle

INPUT Length
INPUT Width

Area ← Length * Width

OUTPUT "The area is ", Area
```

Example 2 – Average test score

- A teacher wants to calculate the average mark achieved on a test amongst students in a class
- The teacher needs to enter how many students in the class and for each students a score out of 50

Input	Process	Output
▪ Number of students ▪ Score per student	▪ TotalScore = TotalScore + score per student ▪ Average = TotalScore / Number of students	▪ Average mark

```
// Calculate the average test score for a class

DECLARE TotalScore : INTEGER
DECLARE Score : INTEGER
DECLARE NumberOfStudents : INTEGER
DECLARE Average : REAL

TotalScore ← 0

INPUT NumberOfStudents

FOR StudentIndex ← 1 TO NumberOfStudents
   OUTPUT "Enter score for student ", StudentIndex
   INPUT Score
   TotalScore ← TotalScore + Score
NEXT StudentIndex

Average ← TotalScore / NumberOfStudents

OUTPUT "The average score is ", Average
```

Programming constructs

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What is sequence?

- Sequence refers to lines of code which are run **one line at a time**
- The lines of code are run in the **order** that they written from the first line of code to the last line of code
- Sequence is **crucial to the flow** of a program, any instructions out of sequence can lead to **unexpected behaviour or errors**

Example

- A simple program to ask a user to input two numbers, number two is subtracted from number one and the result is outputted

Line	Pseudocode
01	INPUT NumberOne
02	INPUT NumberTwo
03	Result ← NumberOne - NumberTwo
04	OUTPUT "The result is ", Result

What is selection?

- Selection is when the **flow of a program is changed,** depending on a **set of conditions**
- The outcome of this condition will then determine **which lines or block of code is run next**
- Selection is used for **validation**, **calculation** and making sense of a **user's choices**
- There are two ways to write selection statements:
  - **if**... **then**... **else**...
  - **case**...

Structure	IF...THEN...ELSE example	CASE statement example
Purpose	Used for **binary decisions** (true/false)	Used for **multiple specific options**
Scenario	Check if a user is old enough to vote	Perform an action based on the direction entered

Pseudocode example	INPUT Age
	IF Age >= 18 THEN
	  OUTPUT "You can vote"
		INPUT Direction
		CASE OF Direction
		  "N" : OUTPUT "North"

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	ELSE
	  OUTPUT "You cannot vote"
	ENDIF
		"S" : OUTPUT "South"
		  "E" : OUTPUT "East"
		  "W" : OUTPUT "West"
		  OTHERWISE : OUTPUT "Invalid direction"
		ENDCASE

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What is iteration?

- Iteration means **repeating a line or block of code** using a **loop**
- It allows a program to perform a task **multiple times** until a condition is met

Types of iteration

Type	Description	Pseudocode Format
Count-controlled	Repeats a fixed number of times	FOR ... TO ... NEXT
Post-condition	Repeats **at least once**, checks the condition **after** running the block	REPEAT ... UNTIL
Pre-condition	Checks the condition **before** running the block	WHILE ... ENDWHILE

```
// Count-controlled
FOR i ← 1 TO 5
  OUTPUT i
NEXT i

// Post-condition
REPEAT
  INPUT Password
UNTIL Password = "Secret123"

// Pre-condition
WHILE Score < 100
  Score ← Score + 10
ENDWHILE
```

Worked Example

A teacher uses a paper-based system to store marks for a class test. The teacher requires a program to assign grades based on these results.

The program will output the grades together with the average mark.

Write a detailed description of the algorithm that will be needed. **[6]**

Answer

Marks awarded for a description of each of the following steps of the algorithm:

1. Reference variables for Count of students and Total marks [1 mark]
2. Loop through all students (Count) [1 mark]
3. Input individual mark (in loop) [1 mark]
4. Compare mark with threshold / boundary values to determine grade (in loop) [1 mark]
5. Output the grade for a student (in loop) [1 mark]
6. Maintain a Total (and Count if required) (in loop) [1 mark]
7. Calculate average by dividing Total by Count and Output (after loop) [1 mark]

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Structured English/Flowchart to Pseudocode

Example – Structured English

A customer is buying an item. If the customer has a discount code, they receive 10% off the price. Otherwise, they pay the full amount. The program should ask for the item price and whether they have a discount code, then calculate and display the final price.

Step 1 – Structured English

1. Ask the user to enter the price of the item
2. Ask the user if they have a discount code
3. If they do have a discount code
   → Calculate 10% off the price
   → Subtract the discount from the original price
4. Otherwise
   → The price stays the same
5. Output the final price

Step 2 – Identifier table

Identifier	Description
ItemPrice	Original price of the item
HasDiscount	TRUE if user has a discount code
DiscountAmount	Amount taken off the original price
FinalPrice	Price after discount is applied

Step 3 – Pseudocode

```
// Ask for item price
INPUT ItemPrice

// Ask if user has a discount code
INPUT HasDiscount

IF HasDiscount = TRUE THEN
   DiscountAmount ← ItemPrice * 0.10
   FinalPrice ← ItemPrice - DiscountAmount
ELSE
   FinalPrice ← ItemPrice
ENDIF
```


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```
// Output the final price
OUTPUT "The final price is ", FinalPrice
```

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Structured English/Pseudocode to Flowchart

Example – Pseudocode

```
INPUT Number

IF Number MOD 2 = 0 THEN
   OUTPUT "Even"
ELSE
   OUTPUT "Odd"
ENDIF
```

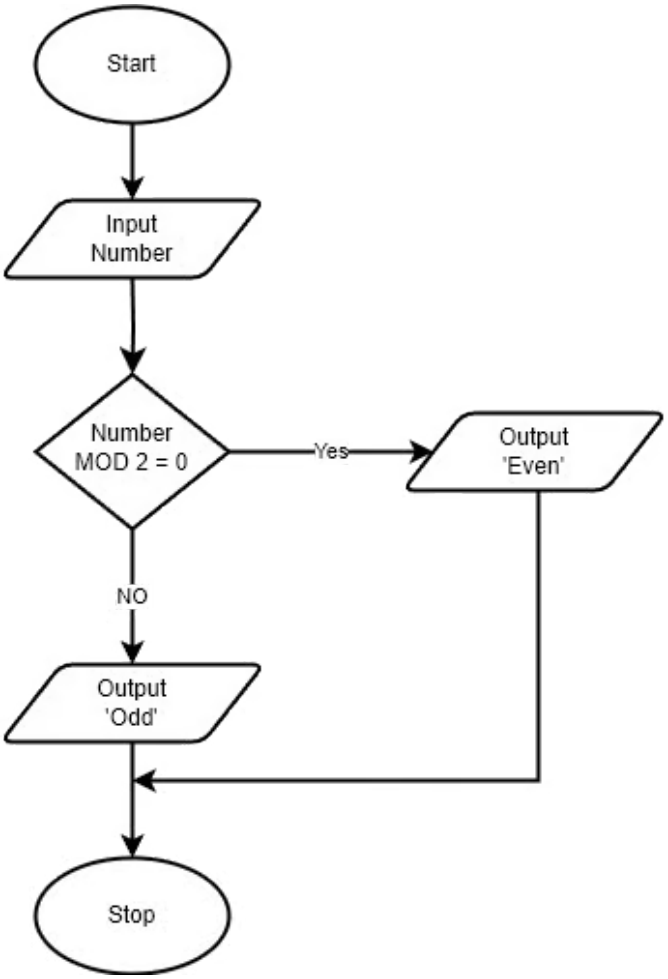
Step 1 – Identifier table

Identifier	Description
Number	The number entered by the user

Step 2 – Flowchart structure

Pseudocode statement	Flowchart symbol	Purpose
INPUT Number	Parallelogram (Input/Output)	User enters a number
IF Number MOD 2 = 0	Diamond (Decision)	Checks if the number is divisible by 2
OUTPUT "Even"	Parallelogram (Input/Output)	Displays message if the condition is true
OUTPUT "Odd"	Parallelogram (Input/Output)	Displays message if the condition is false
Start/End	Oval	Denotes beginning and end of the process
Arrows	Lines with arrows	Indicate flow of control

Step 4 – Flowchart


Start
Input
Number
Number
MOD 2 = 0
Yes
Output
'Even'
NO
Output
'Odd'
Stop

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Stepwise refinement

What is stepwise refinement?

- Stepwise refinement is the process of **breaking down a complex problem** into smaller, more manageable sub-problems in a logical order
- Each sub-problem is **refined step by step** until it is simple enough to be solved with a single subroutine or module
- This method ensures that the overall problem can be solved by addressing each part individually, in a **structured and efficient** way

Relationship to decomposition and top-down design

- **Decomposition** is the general concept of breaking a problem down into smaller parts
- **Top-down design** is the strategy used to perform decomposition
- **Stepwise refinement** is the process used in top-down design to **gradually refine** each major task into simpler sub-tasks
- In other words, **stepwise refinement is how top-down design is implemented**

Benefits of stepwise refinement

- Helps developers **understand and organise** the structure of a program
- Makes **testing and debugging easier** through **unit testing** of individual subroutines
- Encourages **code reuse** by breaking tasks into reusable components
- Supports **collaborative development**, as tasks can be divided between team members
- Each subroutine should:
  - **Be clear and focused** on a single task
  - **Be simple** enough to implement directly
  - **Not need** further breakdown

Example: Calculating student grades

- Here's how stepwise refinement could be applied to a program designed to calculate student grades for a teacher's classes:

Top-level task:

- Calculate grades for all students in all classes

Stepwise refinement:

- **Step 1 – Calculate the grade for each assessment**

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- For each question:
  - Mark the question
  - Store the mark
- Sum the marks for all questions in the assessment

- **Step 2 – Calculate the average grade for each student**
  - Add together grades from all assessments
  - Divide by the number of assessments
  - Store the average
- **Step 3 – Repeat for each class**
  - For every student in the class:
    - Perform Steps 1 and 2
- This structured breakdown allows each task to be turned into a subroutine, simplifying the design, implementation, and testing of the overall program

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